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Ex:4.1

Code :

```
clc; clear; close all;
%% --- Input Parameters ---fp = 1000;
% Passband cutoff (Hz)
fs = 4000;      % Sampling frequency (Hz)
N = 50;         % Filter order (even number preferred) wc = fp/(fs/2); % Normalized cutoff
frequency (0 to 1)
%% --- 1. FIR Low Pass using Rectangular Window ---b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('--- Rectangular Window Coefficients ---'); disp(b_rect);
%% --- 2. FIR Low Pass using Hamming Window ---b_hamm = fir1(N, wc, 'low',
hamming(N+1));
disp('--- Hamming Window Coefficients ---'); disp(b_hamm);
%% --- 3. FIR Low Pass using Hanning Window ---b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---'); disp(b_hann);
%% --- Plot Frequency Responses ---figure; freqz(b_rect,1);
title('Low-pass FIR using Rectangular Window'); figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window'); figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');
```

OUTPUT

```
--- Rectangular Window Coefficients ---0.0126
0
-0.0137
0
0.0150
0
-0.0166
0
0.0185
```

0
-0.0210
0
0.0242
0
-0.0286
0
0.0349
0
-0.0449
0
0.0629
0
-0.1048
0
0.3145
0.4940
0.3145
0
-0.1048
0
0.0629
0
-0.0449
0
0.0349
0
-0.0286
0
0.0242
0

-0.0210

0

0.0185

0

-0.0166

0

0.0150

0

-0.0137

0

0.0126

--- Hamming Window Coefficients ---0.0010 0

-0.0013

0

0.0021

0

-0.0034

0

0.0055

0

-0.0084

0

0.0125

0

-0.0181

0

0.0260

0

-0.0379

0

0.0580

0

-0.1026

0
0.3168
0.4995
0.3168
0
-0.1026
0
0.0580
0
-0.0379

0
0.0260
0
-0.0181
0
0.0125
0
-0.0084
0
0.0055
0
-0.0034
0
0.0021
0
-0.0013
0
0.0010

--- Hanning Window Coefficients ---0.0000 0
-0.0004
0
0.0013

0
-0.0028
0
0.0050
0
-0.0081
0
0.0122
0
-0.0179
0
0.0259
0
-0.0378
0
0.0580
0
-0.1027
0
0.3172
0.5000
0.3172
0
-0.1027
0
0.0580
0
-0.0378
0
0.0259
0
-0.0179
0
0.0122
0

-0.0081

0

0.0050

0

-0.0028

0

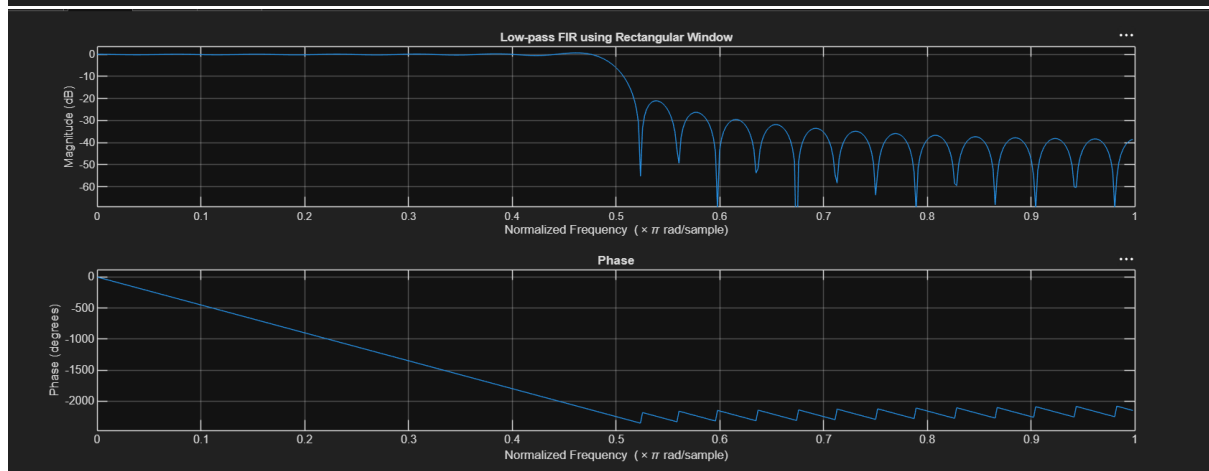
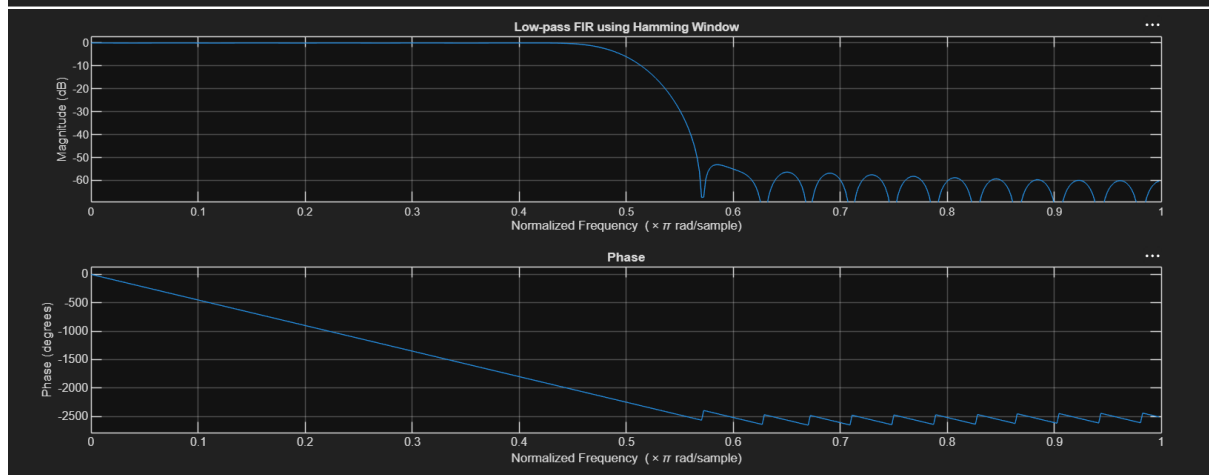
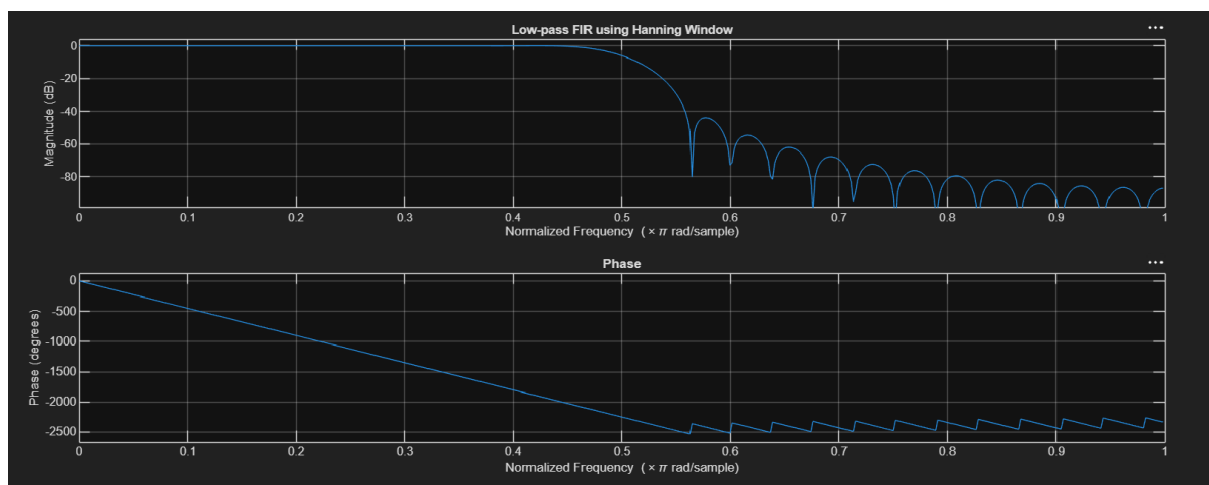
0.0013

0

-0.0004

0

0.0000



Ex :4.2

```
clc; clear; close all;
%% --- Input Parameters ---

fp = 1000;    % Passband cutoff (Hz)

fs = 4000;    % Sampling frequency (Hz)

N = 50;       % Filter order (even number
preferred) wc = fp/(fs/2); % Normalized cutoff
frequency (0 to 1)

%% --- FIR Low Pass using Hamming Window
---b_hamm = fir1(N, wc, 'low', hamming(N+1));

disp('--- Hamming Window Coefficients ---');

disp(b_hamm);

%% --- Plot Frequency Responses

---figure;

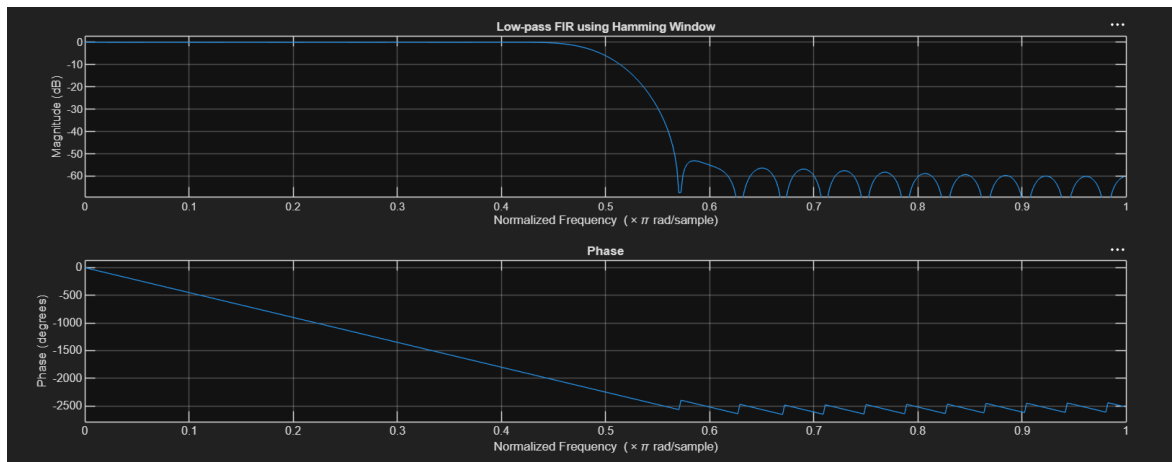
freqz(b_hamm,1);

title('Low-pass FIR using Hamming Window');
```

OUTPUT

```
--- Hamming Window Coefficients
---0.0010
    0
-0.0013
    0
 0.0021
    0
-0.0034
    0
 0.0055
    0
-0.0084
    0
 0.0125
```

0
-0.0181
0
0.0260
0
-0.0379
0
0.0580
0
-0.1026
0
0.3168
0.4995
0.3168
0
-0.1026
0
0.0580
0
-0.0379
0
0.0260
0
-0.0181
0
0.0125
0
-0.0084
0
0.0055
0
-0.0034
0
0.0021
0
-0.0013
0
0.0010



Ex:4.3

PROGRAM

```
clc;
clear;
close
all;

%%% --- Input Parameters ---
fp = 1000;    % Passband cutoff (Hz)
fs = 4000;    % Sampling frequency (Hz)
N = 50;       % Filter order (even number
preferred) wc = fp/(fs/2); % Normalized cutoff
frequency (0 to 1)

%%% --- FIR Low Pass using Hanning Window
---b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann);

%%% --- Plot Frequency Responses
---figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');
```

OUTPUT

-- Hanning Window Coefficients

---0.0000

0

-0.0004

0

0.0013

0

-0.0028

0

0.0050

0

-0.0081

0

0.0122

0

-0.0179

0

0.0259

0

-0.0378

0

0.0580

0

-0.1027

0

0.3172

0.5000

0.3172

0

-0.1027

0

0.0580

0

-0.0378

0

0.0259

0

-0.0179

0

0.0122

0

-0.0081

0

0.0050

0

-0.0028

0

0.0013

0
-0.0004
0
0.0000

